



Density Functional Theory and Conductivity Studies of Boron-Based Anion Receptors

Kevin Leung,^z Mangesh I. Chaudhari, Susan B. Rempe, Kyle R. Fenton, Harry D. Pratt III, Chad L. Staiger, and Ganesan Nagasubramanian

Sandia National Laboratories, Albuquerque, New Mexico 87185, USA

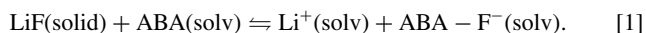
Anion receptors that bind strongly to fluoride anions in organic solvents can help dissolve the lithium fluoride discharge products of primary carbon monofluoride (CFx) batteries, thereby preventing the clogging of cathode surfaces and improving ion conductivity. The receptors are also potentially beneficial to rechargeable lithium ion and lithium air batteries. We apply Density Functional Theory (DFT) to show that an oxalate-based pentafluorophenyl-boron anion receptor binds as strongly, or more strongly, to fluoride anions than many phenyl-boron anion receptors proposed in the literature. Experimental data shows marked improvement in electrolyte conductivity when this oxalate anion receptor is present. The receptor is sufficiently electrophilic that organic solvent molecules compete with F[−] for boron-site binding, and specific solvent effects must be considered when predicting its F[−] affinity. To further illustrate the last point, we also perform computational studies on a geometrically constrained boron ester that exhibits much stronger gas-phase affinity for both F[−] and organic solvent molecules. After accounting for specific solvent effects, however, its net F[−] affinity is about the same as the simple oxalate-based anion receptor. Finally, we propose that LiF dissolution in cyclic carbonate organic solvents, in the absence of anion receptors, is due mostly to the formation of ionic aggregates, not isolated F[−] ions.
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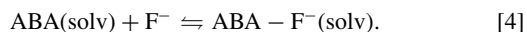
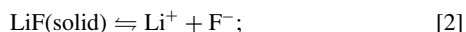
The application of boron-based anion receptors (denoted “ABA” herein) in lithium ion and metal-air batteries has been an area of active research.^{1–10} Experimental and electronic Density Functional Theory (DFT) studies of ABAs have been conducted to examine fluoride anion binding affinity, electrolyte conductivity, redox stability, and other properties critical to battery operations. ABAs are particularly useful in primary carbon monofluoride (CFx) batteries,¹¹ where they improve lithium fluoride (LiF) solubility and prevent this discharge product from clogging the cathode surface. Other proposed benefits of ABAs include F[−] scavenging that improves the solid electrolyte interphase (SEI) in lithium ion batteries, and oxygen anion transportation that can improve lithium-air battery charge/discharge reactions.¹

Boron-based anion receptors often contain strongly electron-withdrawing pentafluorophenyl (−C₆F₅) groups. An often-cited example is tris(pentafluorophenyl)borane (TPFPB).^{2,4} ABAs with multiple phenyl groups tend to be bulky molecules, and can yield viscous electrolytes that impede ionic motion. Slow ionic transport adversely impacts rate capability and other functionalities in an electrochemical storage system. In this work, we focus on an oxalate-based “ABAO” (Figs. 1a and 3a), with the boron atom bound to only one C₆F₅ ring in addition to two oxygen termini of an electron-withdrawing oxalate group.^{1,8} Using electronic structure Density Functional Theory (DFT) techniques, we predict that the ABAO gas phase F[−] binding free energy is comparable to many higher molecular weight anion receptors that have been examined with computational methods.⁷ The gas phase energetics of a subset of ABAs taken from Ref. 7, depicted in Figs. 1 and 3, are re-examined in this work for comparison.

In the presence of liquid solvents, ABAO binding affinity with F[−] is found to remain enhanced compared to other ABA’s. Here we apply DFT methods to examine how different ABA’s and solvent molecules (S) affect the LiF dissolution free energy (Δ*G*_{diss}), according to



The dissolution process can be broken up into steps of a thermodynamic cycle (Fig. 2):



In the equations above, “(soln)” denotes solvation by the organic electrolyte; its absence means the species is in the gas phase. Standard

states (1.0 M concentration) are assumed for Li⁺ and F[−] ions, even when they are considered heuristically to exist in the gas phase. Gas phase contributions to entropy ultimately cancel in Eqs. 2–4 to recover Eq. 1. The free energy change in the first step (Eq. 2) describes solid LiF splitting into ions in the gas phase (Δ*G*_{LiF}). The free energy for solvation of Li⁺ ion (Eq. 3) is quantified by Δ*G*_{solv–Li+}. The free energy in the last step (Eq. 4) describes binding of solvated ABA to F[−] (Δ*G*_{F−}). The LiF solubility or dissolution constant (Eq. 1) in the presence of different ABAs is defined as

$$K_{\text{diss}} = \exp(-\Delta G_{\text{diss}}/k_B T), \quad [5]$$

$$= [\text{Li}^+][\text{F}^-], \quad [6]$$

where Boltzmann’s constant (*k_B*) and the absolute temperature (*T*) define the energy scale, and the dimensionless ion concentrations are normalized by 1.0 M, consistent with the 1.0 M concentration reference systems in our calculations. We also present corroborating

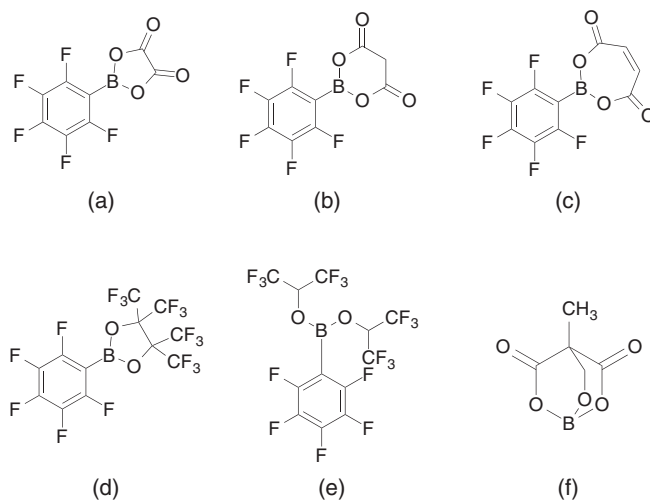


Fig. 1. Structures of (a) oxalic (ABAO), (b) malonic (ABAM), and (c) maleic (ABAE) acid-based ABAs; phenyl boron-based (d) ABA12, and (e) ABA21 proposed in Ref. 7; and (f) ABAT. Here the number in ABAX refers to the ordering used in Ref. 7. Boron atoms are *sp*² hybridized and reside in planar geometries except in ABAT, where B protrudes slightly out of the plane formed by three O atoms. ABA15 of Ref. 7 is similar to ABA12, but with the CF₃ groups replaced by CH₃.

^zE-mail: kleung@sandia.gov

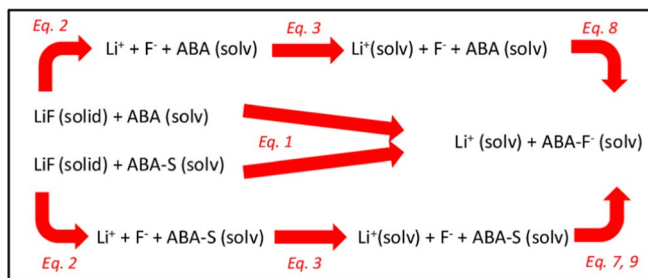


Fig. 2. Thermodynamic cycle schematic. The lower and upper halves pertain to ABA anion receptors that bind or do not bind covalently to a solvent molecule (“S”), respectively. Brackets indicate the phase (solid or solvated); their absence denotes gas phase. Relevant equations in each step are labeled.

experimental data that demonstrate the improvement of electrolyte conductivity when ABAO is present.

Another anion receptor we highlight in our theoretical studies is a recently proposed, geometrically-constrained boron ester.⁵ In traditional ABAs, the B atom exists in a planar, 3-coordinated geometry and exhibits sp^2 hybridization, but becomes sp^3 hybridized when bound to F^- (Fig. 3). The structural changes upon formation of the B-F bond lead to “reorganization energy” penalties (in analogy with electron transfer reactions^{12–14}) that reduce F^- binding affinity (ΔG_{F^-}).¹⁴ By constraining boron in a non-planar geometry even in the absence of F^- , one of the anion receptors of Shanmukaraj et al. (Fig. 1f, henceforth referred to as “ABAT”) is found to exhibit a gas phase F^- binding affinity that exceeds those of planar boron molecules by more than 1 eV (~ 23 kcal/mol or 96 kJ/mol). We only consider the ABAT monomer, not its dimerized/trimerized complexes.

For strong F^- anion receptors like ABAT, we show that it is crucial to include explicit solvent molecules to predict F^- binding affinity. Consider the following possible intermediate steps toward LiF dissolution implicit in Eq. 1:



“S” is a solvent molecule at its liquid density.¹⁵ Eq. 8 (same as Eq. 4) is pertinent to weak anion receptors that do not coordinate to “S.” However, the affinities of ABA toward F^- and solvent molecules tend to be correlated: electrophilic anion receptors that bind strongly to F^- also naturally coordinate to organic solvent molecules with nucleophilic oxygen- and nitrogen-containing terminating groups.¹⁶ Our work will show that Eq. 7 is thermodynamically favored by a substantial amount for ABAO and ABAT. Therefore the exchange reactions of Eq. 9 must be used to predict net F^- binding free energies for these receptors instead of Eq. 8. Such specific solvent binding effects can reduce the selectivity of different ABAs, and can even reverse the ordering of their F^- binding affinity.

Therefore, we also apply DFT methods to survey the interactions of ABAs with four different solvent molecules (Fig. 4): acetonitrile (CH_3CN), dimethyl sulfoxide ($\text{C}_2\text{H}_6\text{SO}$, DMSO), dimethyl carbonate ($\text{C}_3\text{H}_6\text{O}$, DMC), and ethylene carbonate ($\text{C}_3\text{H}_4\text{O}_3$, EC). CH_3CN is a standard solvent used for experimental and computational benchmarking.^{7,17,18} DMSO is used in our synthesis of ABAs (see below). EC and DMC are co-solvents in standard lithium ion battery electrolyte. Even after subtracting large offsetting ABA-S binding free energies, ABAO and ABAT are still predicted to be the most thermodynamically favorable F^- -binding receptors. In terms of kinetics, strong ABA-S interactions may hinder F^- uptake. This is not the focus of our studies. However, we will report experimental evidence of residual ABA-DMSO complexes.

In the theoretical literature, the word “solvent” has been used to describe very different types of solvation models. In the Method section, we distinguish between three treatments of solvation, and emphasize the difference between our solvation model and the one applied in the literature.⁷

Finally, the reorganization energy issue mentioned above is one reason that computational design and/or screening of F^- anion receptor may be challenging unless F^- ions, and in appropriate cases, solvent molecules, are explicitly included in the calculations. We have compared several attributes of ABAO, ABAM, ABAT, ABA12, and ABAE, including several internal angles, their HOMO-LUMO gaps, and dipole moments, prior to binding to F^- or solvent molecules. No clear trend or linear correspondence between these attributes and the F^- binding affinity is discernable. Thus the best way to computationally screen F^- binding affinity is to include F^- explicitly in the calculations.

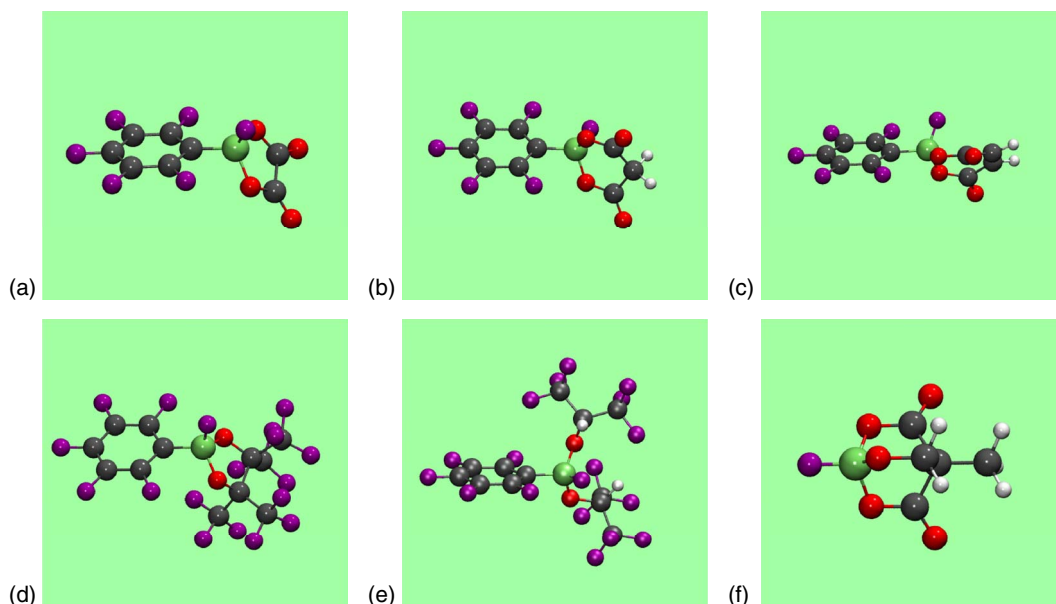


Fig. 3. Optimized structures for (a) ABAO, (b) ABAM, (c) ABAE, (d) ABA12, (e) ABA21, and (f) ABAT, all bonded to an F^- . Boron atoms are sp^3 hybridized in tetrahedral-like geometries. Grey, red, white, lime green, and purple spheres represent C, O, H, B, and F atoms, respectively.

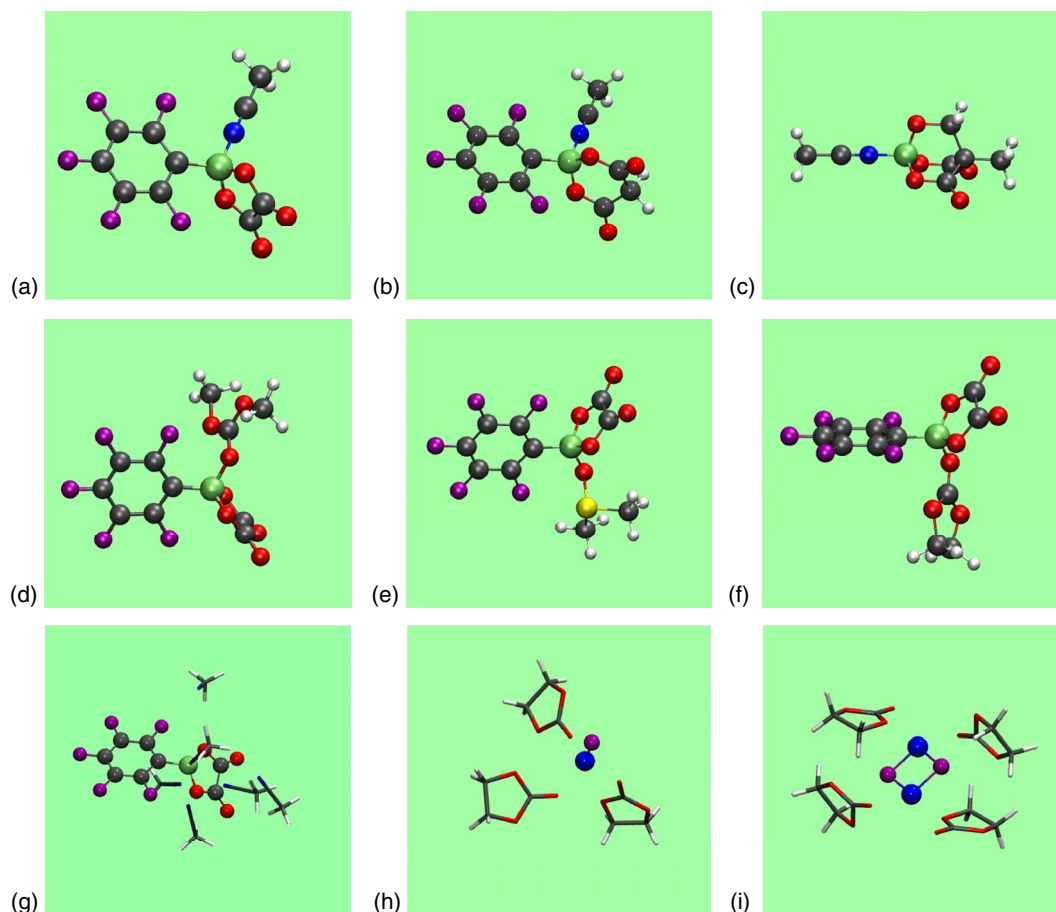


Fig. 4. Optimized structures for (a) ABAO-CH₃CN, (b) ABAM-CH₃CN, (c) ABAT-CH₃CN, (d) ABAO-DMC, (e) ABAO-DMSO, (f) ABAO-EC, (g) ABAO-CH₃CN(CH₃CN)₅, (h) LiF(EC)₃, and (i) Li₂F₂(EC)₄. Grey, red, white, lime green, purple, yellow, blue, and dark blue spheres represent C, O, H, B, F, S, N and Li⁺ atoms, respectively. Fig. 4d depicts DMC in its *cis-trans* conformation, which is most favorable after binding to ABAs, although unbound DMC is most stable as a *cis-cis* conformer.

Method

Gaussian suite of programs.— Most calculations are conducted using DFT with the PBE0 functional.¹⁹ The Gaussian (G09) suite of programs²⁰ and a 6-31+G(d,p) basis are used for geometry optimization of molecular clusters and for computing zero-point-energy (ZPE)/finite temperature corrections. The final, single point energy of each cluster is evaluated using a 6-311++G(3df, 2pd) basis at the optimized geometry.

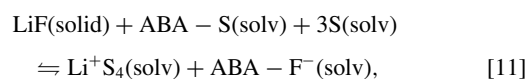
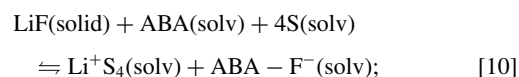
In Eqs. 7, 8, 9, “ABA” can be ABAO, ABAM, ABAE, ABA12, ABA15, ABA21, or ABAT, and the solvent “S” is either CH₃CN, DMC, DMSO, or EC. Here the number in ABAX refers to the ordering used in Ref. 7. When Eq. 8 is used, the ABA boron-site is not bonded to the solvent molecule. In contrast, on the right side of Eq. 7 and the left side of Eq. 9, a B-S chemical bond appears. In general, only the one solvent molecule that bonds covalently with boron appears in each calculation. In the case of CH₃CN, a larger explicit solvation shell is included as a check.

The polarizable continuum model (PCM)²¹ is used to approximate spectator solvent molecules in the outlying bulk electrolyte region surrounding the cluster made up of ABA and solvent molecule(s) in the G09 DFT simulation cell. Various static dielectric constants (ϵ_o) are used to mimic different experimental conditions. Since battery electrolytes typically consist of mixed solvents plus salt, we apply $\epsilon_o = 40.0$ to the outlying region of all these solvent molecules to mimic a uniform, high-dielectric liquid environment. We also consider pure CH₃CN, DMSO, DMC, and EC solvents, with ϵ_o taken to be 46.7, 35.7, 3.1, and 40.0, respectively. EC is a solid and $\epsilon_o \sim 40$ is adopted to reflect a reasonable value for EC/DMC mixtures. Finally, each CH₃CN, DMSO, EC, DMC molecule, and F[−] anion at 1.0 M

concentration, is assumed to occupy a volume of 86.7, 118, 111, 139, and 1668 Å³, respectively. These values are deduced from densities/concentrations at room temperature. In the case of EC, the value comes from the solvent density at higher temperature. The volumes lead to small, < 0.16 eV modifications of default translational entropies reported by the G09 software for pressure equal to 1 atm. standard state reaction gas phase conditions.

Different organic solvents solvate Li⁺ to different extents (Eq. 3), and Li⁺ solvation free energies are calculated using Li⁺S₄ clusters, with “S”=CH₃CN, EC, DMSO, and DMC. Four explicit solvent molecules are included because Li⁺ is generally 4-coordinated in polar solvents.^{17,22–27} In all cases, we report results associated with the bare, unsolvated F[−]. These unsolvated F[−] ions appear only in intermediate steps in the calculations, not the final result (Eq. 1). The anion is expected to exist either as LiF solid or ABA-F. The one instance where the free energy of solvated F[−] is needed is when predicting LiF solid solubility in liquid electrolyte in the absence of ABAs, for comparison purposes. Even in this case, we find that F[−] should exist as (Li⁺)_n/(F[−])_n aggregates (e.g., Fig. 4h and Fig. 4i).

Putting these considerations together, the total LiF dissolution reaction free energies (ΔG_{diss}) in Eq. 1 are calculated using either of the following equations,



depending on whether the ABA in question forms a thermodynamically stable complex with the solvent “S.”

To assess basis set superposition effects (BSSE) when using the 6-311++G(3df,2pd) basis, standard BSSE corrections are calculated for four gas reactions: $\text{ABAO} + \text{EC} \rightarrow \text{ABAO-EC}$, $\text{ABAM} + \text{EC} \rightarrow \text{ABAM-EC}$, $\text{ABAO} + \text{F}^- \rightarrow \text{ABAO-F}^-$, and $\text{ABAM} + \text{F}^- \rightarrow \text{ABAM-F}^-$. The values are 0.07, 0.07, 0.15, and 0.15 eV, respectively. Thus BSSE does not appear to change the preference for EC and F^- binding among different ABAs. The maximum impact on net ABAO-F^- binding, via Eq. 9, is 0.08 eV, or $\sim 3 k_B T$. Since it has been suggested BSSE calculations can be overestimated,^{29,28} and such calculations cannot be performed in the presence of a dielectric environment (PCM solvation method, see below, which should contract the F^- electron cloud and reduce BSSE), we have not conducted BSSE calculations for other ABA's and solvent molecules or included BSSE in the Results section.

Different types of solvation models.— It is important to distinguish explicit versus implicit solvent treatments in atomistic length-scale simulations. Most electronic structure (e.g., quantum chemistry or DFT) calculations involve localized basis sets, and a small molecular cluster representing the chemical reaction zone. The cluster is relaxed to its most stable atomic configuration as though it is at zero temperature ($T = 0$ K). The effect of a finite temperature is typically approximated, post-processing, using a harmonic expansion to account for vibrational motion³⁰ and by adding translational/rotational entropies. The outlying region containing liquid solvent is treated implicitly, using dielectric continuum (solvation reaction field) methods.²¹ If the cluster or reaction zone contains no explicit solvent molecule, the solvation treatment is henceforth described as “type 1.” If at least one solvent molecule is included, it is dubbed “type 2.”

A more costly approach, which in principle involves fewer approximations, is ab initio molecular dynamics (AIMD, also known as DFT-MD).³¹ All atoms, including solute and solvents, are treated at the same DFT level, and periodic boundary conditions are generally applied. The simulation is conducted at finite temperature via solving Newton's equation of motion. This approach (“type 3”) avoids the arbitrary demarcation of explicit and implicit solvent regions, and is in principle exact given sufficiently large simulation cells, long simulation times, and accurate DFT functionals. In contrast, while more solvent molecules usually give improved results with type 2 solvation,³² this is not guaranteed. In the limit of a large number of solvent molecules, the nature of the geometry optimization used with type 2 means that the most stable state of the system should be a crystallized solid of solvent molecules embedding the solute as an impurity. This clearly does not describe a liquid state configuration. Furthermore, type 2 calculations typically treat finite temperature effects via harmonic expansions even when the pertinent solvent motion is diffusive. This approximation can exaggerate the contribution of zero-point energies. An alternative to DFT-based AIMD is to conduct molecular dynamics using classical force fields, where the accuracy of predictions depends on the quality of the fitted force fields.^{17,33,34} Other solvent-specific errors associated with continuum solvation techniques have also been discussed.¹²

In this work, we will only consider type 1 and type 2 solvation. In most cases, one explicit solvent molecule is included. This is reasonable because only one solvent molecule can covalently bond to the boron site. Solvent binding causes extensive geometric changes in most ABAs (Fig. 4). In the ABA literature, to our knowledge, type 1 solvation has been used exclusively.⁷ Even such a *purely* continuum approach (type 1) has predicted that the solvent reduces the differential affinity for F^- ($\Delta\Delta G_{\text{F}^-}$) among different ABA's, defined as the ΔG_{F^-} change when switching from one ABA to another (Eq. 4), by approximately 25% compared to gas phase $\Delta\Delta G_{\text{F}^-}$.⁶ However, type 1 solvation does not yield the correct geometry changes in ABA (Fig. 4). The present work shows that including an explicit solvent molecule is crucial for strong anion receptors, and can lead to a large modification of F^- -binding free energy.

VASP calculations.— To compute the zero temperature total energy of LiF solid, we apply the VASP code,^{35,36} PAW pseudopotentials,³⁷ and PBE0 functional¹⁹ in plane-wave-based DFT calculations carried out in periodic boundary conditions. The commonly used hybrid PBE0 functional is chosen because it is implemented in both G09 and in VASP. An energy cutoff of 500 eV for plane waves and a 10^{-5} eV wavefunction convergence criterion are enforced. The optimal lattice constant (4.02 Å) and cohesive energies are calculated in a face-centered cubic cell with a 2-atom unit cell and $4 \times 4 \times 4$ Monkhorst-Pack Brillouin sampling. LiF phonon dispersions are then computed to estimate finite temperature corrections in the harmonic approximation. Phonon calculations apply the same settings, except that a 512-atom (16.08 Å)³ supercell with Γ -point sampling is applied and the less expensive PBE functional is used for this larger simulation cell.³⁸ A finite difference approach is applied to calculate vibrational force constants. This yields the dynamical matrix, the eigenvalues of which are vibrational frequencies (ω).³⁹ The vibrational correction to the free energy is

$$\Delta G_{\text{harm}} = 1/\beta \sum_i \int_{\mathbf{k}} d\mathbf{k} \log[2\sinh(\beta \hbar \omega_{\mathbf{k},i}/2)], \quad [12]$$

where β is the inverse thermal energy ($1/k_B T$), $\hbar$ is Planck's constant, $\{\mathbf{k}\}$ spans the Brillouin zone, and i is the composite index for the 6 eigenvalues ω_{ki} of the dynamical matrix at each $\mathbf{k}$ -point. Eq. 12 yields a small, 0.067 eV thermal contribution. Therefore, we have not pursued improvement to the phonon calculation, e.g., via using the more accurate hybrid PBE0 functional.

Experimental method.— Anion binding agents were synthesized using previously reported methods⁴⁰ and tested for electrochemical performance. The anion receptors considered include ABAO (IUPAC name: 2-(perfluorophenyl)-1,3,2-dioxaborolane-4,6-dione), ABAM (2-(perfluorophenyl)-1,3,2-dioxaborinane-4,6-dione), and a pinacol-based ABA (4,4,5,5-tetramethyl-2-(perfluorophenyl)-1,3,2-dioxaborolane) equivalent to ABA15 considered in Ref. 7. One additional step was executed to remove DMSO, used in synthesis, from the resulting products. All solids were redissolved in acetone with an excess of LiF. Undissolved LiF was removed by syringe filtration (2 μm) and the filtrate condensed by slow evaporation in air.

Electrochemical cells were assembled using 2032 coin cells, which utilized stainless steel electrodes separated with a polyimide spacer to ensure a uniform electrode separation. Electrolytes were composed of 3:7 (wt%) ethylene carbonate/ethyl methyl carbonate (EC:EMC) and 1.0 M ABA. Equivalents of 1.0 M LiF crystals were added to each electrolyte solution. All solutions were stirred thoroughly, but the total LiF content in solution was variable based upon the efficiency of fluoride binding by the ABA molecules. Lower affinity binding agents resulted in solutions that were much lower concentration than 1 M after filtration. Approximately 1 mL of electrolyte for each of the tested binding agents was flooded into the coin cell prior to sealing the cell shut. This was done to ensure that there was complete flooding of the electrochemically accessible area for both electrodes. Conductivity was determined using a Solartron 1287 and 1260 stack by measuring the AC impedance in the frequency range of 100 kHz to 0.1 Hz. The peak-to-peak voltage of the AC signal was limited to 5 mV to avoid any distortion in the response.

X-ray single-crystal data collection was conducted using a Bruker-APEX/CCD diffractometer (Mo $K\alpha$, $\lambda = 0.71073$ Å). Indexing and frame integration were performed using the APEX-II software suite. Absorption correction was performed using SADABS (numerical method) also within the APEX-II software. The structures were solved and refined using SHELXS-97 contained in SHELXTL v6.10 packages.

Results

Oxalate and boron ester ABA's are good anion receptors.— Table 1 lists new predictions for ABAO, ABAT, and re-examines several

Table I. F^- and CH_3CN (“S”) binding free energies with various ABAs computed using Eqs. 7–9, in units of eV (~ 96 kJ/mol). No explicit solvent is present except in the last two rows (ABA-S and ABA-F $^-$), where one CH_3CN is coordinated to the ABA boron site and Eq. 9 (instead of Eq. 8) is used to compute F^- binding free energy. ABA12 and ABA15 fail to bind to CH_3CN in the calculations.

	ABAO	ABAM	ABAE	ABA12	ABA15	ABA21	ABAT
ABA- F^- ($\epsilon_o=1$)	−4.396	−4.171	−4.128	−3.741	−2.679	−4.234	−5.411
ABA- F^- ($\epsilon_o=40$)	−5.962	−5.650	−5.660	−5.450	−4.566	−5.581	−7.238
ABA-S ($\epsilon_o=40$)	−0.249	−0.002	−0.069	NA	NA	−0.034	−1.533
ABA- F^- *($\epsilon_o=40$)	−5.714	−5.648	−5.591	−5.405	−4.521	−5.547	−5.705

fluoride receptors explored in Ref. 7. First we discuss their F^- -binding free energies either in vacuum ($\epsilon_o = 1$), or with type I solvation using a polarizable dielectric continuum (PCM) implicit solvent model (Eq. 8). The gas phase binding *enthalpy* (not shown) are comparable to those reported in Ref. 6, although a somewhat different basis set is used compared with that work so as to be compatible with methods used for ABAO and ABAT herein. For a first estimation of solvation effects, ϵ_o is set to 40 to mimic a generic high dielectric liquid environment. As discussed above, for the purpose of this calculation, F^- is not solvated, and its energetic contribution is constant for all ABAs and solvents.

In vacuum (first row of Table I), ABAO binds more strongly to F^- than almost all other ABAXs, even those with multiple $-C_6F_5$ electron-withdrawing groups. The exception is ABAT, which is by far the most fluorophilic. As discussed in the Introduction, ABAT alone has its boron atom in a non-planar geometry in its F^- -free state and is less adversely affected by the reorganization energy cost when binding F^- . When *only* the solvent dielectric continuum is added (second row of Table I), the ΔG_{F^-} ordering remains largely unchanged. It is of interest to compare ABA12 and ABA15, which differ only by their $-CF_3$ and $-CH_3$ groups. The electron-withdrawing $-CF_3$ groups stabilizes of F^- binding by 0.884 eV (20.4 kcal/mol) over ABA15.

Including explicit CH_3CN solvent molecule(s).— Free energies computed using Eq. 8 may overestimate F^- -binding in polar solvents because the boron site may bond to solvent molecules. Next we examine the effect of an explicit CH_3CN solvent molecule coordinated to these ABA's (last two rows of Table I). Figs. 4a, b, c depict the optimized, most enthalpically favorable geometries of ABA- CH_3CN at $\epsilon_o = 40.0$. The N-atom terminus of the solvent coordinates to the boron site, just like F^- (Fig. 3), leading to sp^3 -hybridization of the B-atom and significant distortion of the molecular geometries compared with unbound ones (Fig. 1a and 1b). Such distortions are not observed when type I solvation treatment is used.

The solvent coordination reactions, $ABAO+CH_3CN \rightleftharpoons ABAO-CH_3CN$ and $ABAM+CH_3CN \rightleftharpoons ABAM-CH_3CN$, exhibit free energy changes of -0.249 and -0.002 eV, respectively. CH_3CN binding to ABAO is therefore exothermic while it is almost thermoneutral for ABAM. Note that the zero temperature binding *enthalpies* to CH_3CN are favorable in both cases: -0.642 and -0.417 eV for ABAO and ABAM, respectively. As is typical of $A + B \rightleftharpoons C$ reactions, the translational and rotational entropy penalties add up to more than 0.4 eV. They negate, or almost negate, the substantial favorable reaction enthalpies. ABAE and ABA21 also exhibit small CH_3CN binding free energies.

Since ABAO binds favorably to CH_3CN , Eq. 9 should be used to examine F^- binding affinity. With the explicit solvent contribution added, the F^- -affinity (ΔG_{F^-}) of ABAO lessens from -5.965 eV to -5.714 eV. For ABAM, ΔG_{F^-} is almost unmodified, becoming -5.648 eV (Table I). Inclusion of explicit solvent therefore reduces the differential F^- affinity ($\Delta \Delta G_{F^-}$) between ABAO and ABAM from ~ 0.31 eV to ~ 0.07 eV. Even so, ABAO remains a slightly better F^- receptor. The F^- -binding affinity of ABAE and ABA21 are slightly reduced when an explicit CH_3CN is included in the calculations. ABA12 and ABA15 do not bind to CH_3CN —even metastable ABA12- CH_3CN and ABA15- CH_3CN structures cannot be optimized with

$\epsilon_o = 40$ in the calculations. Hence Eq. 8 should be used instead, and their ΔG_{F^-} remain unchanged; an explicit CH_3CN should not be present in these models.

ABAT binds strongly to acetonitrile. The free energy associated with Eq. 7 is -1.533 eV (-35 kcal/mol). If an explicit CH_3CN is not used in the calculations, the F^- -binding affinity would be overestimated by this amount. This translates into a 6×10^{24} fold error in the equilibrium dissolution constant (K_{diss} , Eq. 5). Subtracting Eq. 7 from Eq. 8 yields Eq. 9, from which ΔG_{F^-} drastically drops from -7.238 eV to -5.705 eV. With this significant modification due to explicit solvent effects, the F^- binding affinities of ABAO and ABAT in CH_3CN become almost identical, despite the large gas phase disparity.

We have also considered adding more solvent molecules to evaluate Eq. 8. Figure 4g depicts six CH_3CN in the first solvation shell of ABAO, in addition to the use of the PCM dielectric surrounding the explicit solvent region. The F^- binding *enthalpies*, without vibrational corrections, are predicted to be -5.620 and -5.570 eV with one and six CH_3CN , respectively. They are within 50 meV of each other. When thermal and zero-point corrections are included, a slightly larger 80 meV difference is predicted. This is likely because the additional CH_3CN are only weakly coordinated to the ABAO- CH_3CN complex, yet the ABA-solvent interactions are still treated as harmonic vibrational modes. This approximation can lead to overestimated zero-point energy corrections that do not completely cancel in the reactants and products. Nevertheless, the predictions are sufficiently similar with 1 and 6 solvent molecules that we only consider one explicit molecule in all following calculations.

Survey of other solvent molecules, with ABAO, ABAM, and ABAT.— DMSO is used during our synthesis of ABAO, and it exhibits the highest ABA-binding free energies among solvents examined in this work (Table II). Figure 5 depicts the X-ray crystal structure of ABAO-DMSO prior to solvent exchange, and after DMSO is replaced by F^- . The before-exchange (left panel) reconstruction clearly demonstrates that the boron site becomes sp^3 hybridized due to formation of a covalent bond with the oxygen site of DMSO. This hybridization remains after F^- substitutes for DMSO. There are also some extra materials not expected for this reconstruction post DMSO solvent-exchange (right panel). This is due to residual material that remains within the crystal structure during the evaluation. The software assigns lithium to the residual electron density remaining within the sample. The X-ray reconstruction in Fig. 5 is sufficient to demonstrate the definitive change from DMSO-boron interaction to fluorine-boron interaction when an aggressive solvent exchange is conducted. Without this exchange, the primary reaction site for these ABA would be bound to solvents and unavailable for use to scavenge LiF during battery operation.

The DMSO-bound ABAO structure predicted from DFT (Fig. 4e) looks similar to the left panel of Fig. 5, except that the DMSO molecule is rotated so that one of its $-CH_3$ protons coordinates to an F^- on the phenyl ring. Since the calculation only contains one ABAO-DMSO complex, and omits explicit representation of the surrounding molecules to which DMSO can coordinate in the X-ray spectrum sample, the difference is understandable. The rotational conformational difference is not expected to lead to a significant change in the predicted energy.

Table II. F^- (ABA-F) and solvent-binding (ABA-S) free energies, Li^+ solvation free energies computed with an explicit solvent shell of 4 molecules (Li^+S_4), and sum of these two (ABA-F/ Li^+), in eV units (~ 96 kJ/mol). In the case of F^- , the asterisk refers to the corrected binding free energies (Eq. 9) if the ABA binds to the solvent molecule (i.e., Eq. 8 yields an attractive free energy). The ABA15-F/ Li^+ value in EC solvent ($\epsilon_o = 40$) is -8.926 eV. The *cis-trans* DMC conformation is most favorable when bound to ABA's.

	CH ₃ CN	DMSO	DMC	EC
ϵ_o	35.7	46.7	3.1	40.0
ABAO-F	-5.958	-5.968	-5.496	-5.962
ABAM-F	-5.637	-5.653	-5.215	-5.650
ABAT-F	-7.232	-7.245	-6.672	-7.238
ABAO-S	-0.248	-0.686	-0.087	-0.114
ABAM-S	-0.001	-0.414	-0.139	+0.083
ABAT-S	-1.532	-2.059	-1.346	-1.450
ABAO-F*	-5.710	-5.282	-5.409	-5.848
ABAM-F*	-5.636	-5.239	-5.076	-5.650
ABAT-F*	-5.700	-5.186	-5.326	-5.788
Li^+S_4	-4.746	-5.008	-2.195	-4.445
ABAO-F/ Li^+	-10.456	-10.290	-7.604	-10.293
ABAM-F/ Li^+	-10.382	-10.247	-7.271	-10.095
ABAT-F/ Li^+	-10.446	-10.194	-7.521	-10.223

Table II shows that, at room temperature, both ABAO-DMSO and ABAM-DMSO complexes are favorable. Even after subtracting the free energy cost of breaking the ABA-DMSO bond to form ABA-F (i.e., using Eq. 9 rather than Eq. 8), ABAO retains a slight preference for F^- -binding relative to ABAM in DMSO, while ABAT is slightly inferior to ABAO by ~ 0.1 eV.

Other solvents like DMC and EC have smaller specific solvent effects than DMSO (Table II). EC exhibits an unfavorable binding free energy with ABAM. Therefore ABAM should retain its planar geometry, and type 1 dielectric continuum (Eq. 10) calculations suffice for this ABA. In contrast, ABAO binds to all solvents. So does ABAT, which exhibits much larger binding free energies than ABAO. After subtracting the solvent-coordination effects, ABAT is predicted to be only slightly inferior to ABAO for binding- F^- .

LiF ionization and Li^+ solvation free energies.— Using the VASP code, the free energy of splitting LiF solid into Li^+ and F^- ions in the gas phase (ΔG_{LiF} , Eq. 2) is found to be 10.098 eV per formula unit. This includes finite temperature corrections due to solid state vibrational motion at $T=300$ K (Eq. 12) and the translational entropy gained by Li^+ and F^- corrected to 1.0 M concentration, even though the ions are assumed to be in gas phase in calculations associated with Eq. 2. Gas phase entropic contributions cancel in the final results. Li^+ solvation free energies are also needed in dissolution predictions. With Li^+S_4 clusters and the PCM dielectric continuum approximation outside the cluster, Li^+ solvation free energies ($\Delta G_{solv-Li^+}$, Eq. 3) in CH₃CN, DMSO, DMC, and EC are predicted to be -4.746 , -5.008 , -2.195 , and -4.445 eV, respectively, when computed at the respective ϵ_o of the pure solvent (Table II). The EC result is consistent with earlier predictions of gas phase energy changes for LiF splitting and Li solvation.⁴³ Note that the DMC solvation value may be underestimated by a small fraction of an electron-volt due to the neglect of the significant quadrupolar solvent contributions in the PCM dielectric continuum treatment, which is specific to this molecule.^{12,44} However, F^- -binding to ABA in DMC is so much less favorable than in other solvents (Table II) that this small solvation error should not affect the qualitative conclusions below.

LiF solubility predictions.— The dissolution free energies (ΔG_{diss}) according to Eq. 10 or 11 can now be calculated by adding ΔG_{F^-} , $\Delta G_{solv-Li^+}$ (the last rows of Table II), and $\Delta G_{LiF} = 10.098$ eV. A negative ΔG_{diss} means that LiF dissolution is thermodynamically favorable. From Table II, we predict that LiF should be mostly soluble in 1.0 M ABAO, ABAM, and ABAT in the presence of CH₃CN, DMSO, and EC. The exceptions are ABAM in EC and ABAT in EC and DMSO, where the solubility estimates are within the margin of computational uncertainties. Even though we have not applied the most advanced DFT functionals,^{45,43} the uncertainty of the computational method is unlikely to exceed a few tenths of an eV, while $\Delta \Delta G_{diss}$'s between different ABA's should be much more reliable than that.

As discussed in Refs. 33 and 34, it is challenging to relate the predicted dissolution enthalpy (or free energy) to absolute solubility. Here the solubility for a binary compound like LiF is estimated via $(K_{diss})^{0.5}$ (Eq. 6). We focus on EC molecules because they are the closest approximation to the electrolyte used in our experiments (an EC/EMC mixture) among the solvents examined in our modeling

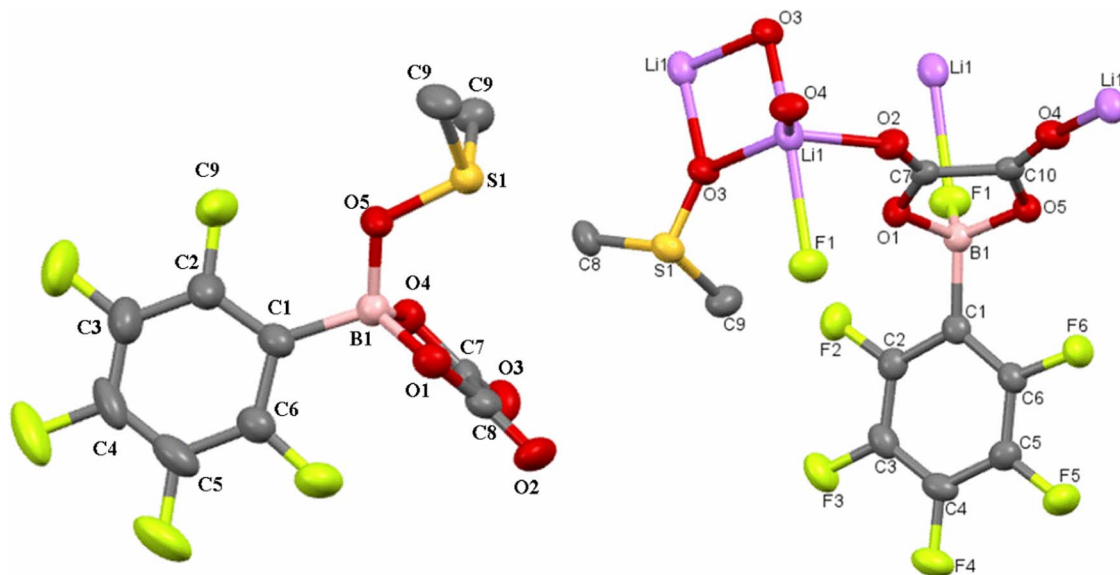
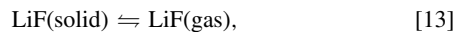


Fig. 5. X-ray crystal structures of ABAO-DMSO prior to solvent exchange (left panel) and ABAO- F^- after solvent exchange (right panel). The color scheme used is slightly different from Figs. 3-4. F, B, and Li are in green, light pink, and dark pink instead of purple, dark green, and dark blue. Protons are subsumed into carbon atoms.

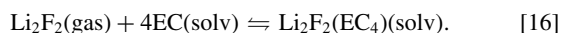
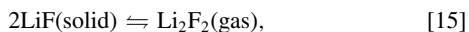
work. For ABAO-F⁻/EC, Eq. 11 yields $K_{\text{diss}} = \exp(-(-10.293 + 10.098)\text{eV}/k_{\text{B}}T)$. This translates into a 43.3 M solubility. The value is unphysically high because the calculations assume infinite Li⁺ and ABAO-F⁻ dilution (i.e., ideal solution with unit solute activity) and neglect steric and electrostatic repulsion between like-charged ions. Inherent approximations in DFT functionals may also contribute to the extremely high value. Nevertheless, the prediction of high LiF solubility in ABAO is consistent with the measured high mobility (Fig. 6 below). With ABAM-F⁻, the solubility is predicted to be a much lower 0.94 M. This value may again be overestimated. However, the trend that LiF solubility decreases going from ABAO to ABAM is qualitatively consistent with the reduced ABAM conductivity.

We have also considered LiF solubility in ABA15 (not listed in Table II). $\Delta G_{\text{solv-Li}^+}$ and ΔG_{F^-} sum to -8.966 eV in the presence of ABA15 with the EC-appropriate $\epsilon_o = 40$ for type 1 solvation. The ABA15-assisted Li⁺/F⁻ solubility in EC is predicted to be 1.5×10^{-10} M. Hence ABA15 is not expected to contribute to dissolution of LiF solid. The measured LiF solubility and conductivity in ABA15 may in fact reflect properties in the absence of any ABA.

LiF solubility in the absence of ABA is therefore needed for comparison purposes. Here we find that, if we treat Li⁺ and F⁻ as well-separated solvated species surrounded by a dielectric continuum, the predicted solubility is a miniscule, unphysical 10^{-16} . This indirectly suggests that Li⁺ and F⁻ must form ionic aggregates. We consider the two smallest charge-neutral aggregates, a Li⁺/F⁻ contact ion pair (Fig. 4h) and a Li₂F₂ cluster (Fig. 4i). Three (3) and four (4) EC molecules are added to these clusters, respectively, to keep all Li⁺ 4-coordinated to either EC or F⁻. The LiF dissolution reactions in the absence of ABAs can be written as the respective cycles



and



Solid state sublimation energies and liquid state solvation free energies are computed using the VASP and G09 codes. The resulting ΔG_{diss} for formation of a LiF contact ion pair is predicted to be +0.986 eV, which translates into a 5.3×10^{-9} M effective LiF solubility. This ion-pair solubility value is already larger than that computed for Li⁺ and F⁻ solvation via formation of ABA15-F⁻ complexes. For Li₂F₂, ΔG_{diss} is +1.444 eV, which yields a 8.7×10^{-7} M solubility when (K_{diss})^{0.25} is used to represent [Li⁺] and [F⁻] solubility. This is orders of magnitude higher than the contact ion pair value. The overall trend suggests that ion aggregates may be responsible for much of the LiF solubility in the absence of ABAs. We will return to this point in the next subsection.

Using molecular dynamics simulations based on classical force fields, Tasaki *et al.* also predicted unfavorable heats of solvation for LiF solid in DMC and EC solvents in the absence of ABAs.^{33,34} A direct comparison between these predictions and our results is difficult because different reference states are used. Unlike Li⁺ and F⁻, bulky ABA molecules are not likely to form aggregates, except for the constrained boron ester ABAT.⁵

Conductivity measurements, experimental estimates of solubility, and comparison to DFT modeling.— Figure 6 depicts the electrolyte conductivity measured in the coin cells as a function of temperature. ABAO exhibits the highest conductivity at all temperatures, while ABAM is a factor of 2-3 lower. The conductivity in ABA15 is negligible. Electrolyte conductivity should be strongly correlated with the ability of the electrolyte to dissolve LiF. The observed conductivity trend is therefore qualitatively consistent with the predicted ABAO>ABAM>ABA15 LiF solubility discussed in the last section.

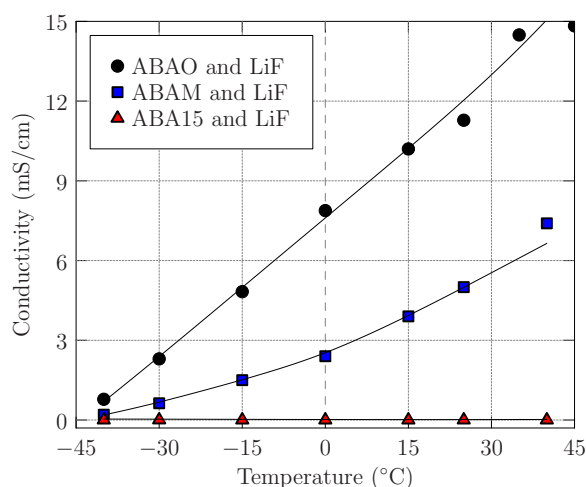


Fig. 6. Electrolyte conductivity. Black, blue, and red represent 1.0 M ABAO, ABAM, and ABA15 added to the electrolyte (30:70 weight % EC/DMC), respectively.

To make a semi-quantitative estimate of the amount of lithium in solution when ABA15 is present, it was assumed that the conductivity varied with concentration, similar to other battery electrolyte systems.^{41,42} The conductivity of the ABAO solutions were assumed to be maximized for the purpose of this analysis (i.e., all LiF added was dissolved, yielding a 1.0 M concentration). Also, ABAO was used as a reference for the ABA15 solution. Conductivities were then approximated using a fourth order polynomial fit.⁴¹ The concentration of dissolved LiF that was calculated in solution using this evaluation for the ABA15 solution was approximately 5×10^{-4} mol/L. This represents a large reduction in overall fluoride binding affinity for ABA15 over the other demonstrated binding agents. As mentioned above, our estimated 5×10^{-4} mol/L LiF solubility in the ABA15 solution is likely due to Li⁺ and F⁻ ions and their ionic aggregates, and not due to ABA15-F⁻ complexation.

Tasaki *et al.* reported that the LiF concentration in neat DMC is $1.7 \pm 0.4 \times 10^{-4}$ mol/L, or 4 ± 1 ppm after filtration through 2 μm filter.³³ This is slightly lower but is of the same order of magnitude as the 5×10^{-4} mol/L solubility we estimate for EC/DMC/apparently-inert ABA15. EC/DMC has a higher dielectric constant and is expected to be a better solvent for LiF dissolution. Hence their measurements and ours are broadly consistent with each other.

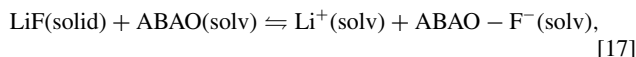
Jones *et al.* have shown that LiF solubility in the absence of ABAs can strongly depend on whether filtration was performed, and on the filtering pore sizes.⁴⁶ Without filtering, LiF solubility in DMC was reported at 2.2×10^{-2} mol/L,⁴⁶ more than 100 times higher than post-filtering.³³ Large pore sizes in filtering crucible also increase the apparent LiF solubility.⁴⁶ The solubility of LiF in EC or EC mixtures was only reported without filtering,⁴⁶ and a direct comparison with filtered DMC/LiF solutions cannot be made.

The study of Jones *et al.* appears to confirm the role of LiF ionic aggregates in LiF solubility suggested by our calculations. As discussed above, LiF solubility increases as the end product varies from isolated Li⁺ and F⁻, to LiF pairs, to Li₂F₂ clusters, although there remains a substantial difference between the cluster sizes considered in theory and experimental pore sizes. Quantitatively, the DFT-predicted LiF solubility of 8.7×10^{-7} M remains substantially below the experimental estimate of $\sim 5 \times 10^{-4}$ M. However, the discrepancy is expected to shrink as larger clusters are considered in DFT calculations. In the future, it is of interest to estimate the LiF solubility in the absence of ABA using nanofiltration-prepared electrolytes in experiments, and at the same time perform simulations of higher order aggregates (LiF)_n, $n > 2$, which are more costly computationally, to further confirm this point.

Conclusions

Using both DFT predictions and conductivity measurements, we have shown that the oxalate-based ABA (“ABAO”) has LiF-solvation properties that makes it a promising fluoride receptor for primary carbon monofluoride (CFx) batteries. The suitability of this anion receptor in other batteries like lithium-ion or lithium-air may depend on factors other than LiF conductivity,⁶ such as its interaction with PF₆[−] which is not present in primary batteries. However, the solvent effects emphasized in this work has general interest in liquid electrolyte-based energy storage devices beyond CFx.

After accounting for explicit solvent effects, the equilibrium constant K_{diss} for the reaction with ABAO,



is the largest among the ABAs tested here. The trend is confirmed by explicitly comparing the conductivity of three ABAs we have synthesized and tested. The predicted LiF solubility in ABAO is at least comparable to that of a recently proposed ABAT with the boron atom in a non-planar environment,⁵ after specific solvent effects are taken into account. Indeed, for these strong F[−]–binding anion receptors, we find that including explicit ABA-S covalent bonding for different choices of solvent is crucial. Omitting the solvent molecule in the calculations can lead to ABA-F[−] binding coefficients that are in error by many orders of magnitude. Our results therefore emphasize the importance of considering explicit F[−] and solvent-molecule binding in calculations when conducting computational design/screening of anion receptors. In the absence of ABAs, we propose that LiF dissolves in cyclic carbonate organic solvents mostly through the formation of ionic aggregates; isolated and well-solvated Li⁺ and F[−] species likely exist at low concentrations in LiF solutions.

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